Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Assembly Lines**
b. Team leader name: **Harshvardhan Singh**
c. Problem Statement: **Sustainability & Social Impact. Relay gets help to people when cyclones take down the cell network.**

Brief about the idea

Relay carries a cry for help across phones with no signal, and brings the answer back.

After a cyclone, help exists. It just can't find the people who need it. The requester has no bars; the duty officer has boats and volunteers but no idea which village to send them to first.

Relay turns a voice note into a ~100-byte ticket on the phone with Gemma, offline. The ticket hops phone to phone over a Bluetooth mesh until any phone finds one bar of signal. Gemini agents in the cloud merge, locate and prioritise it, and the district control room approves a dispatch in one click. The confirmation rides the same hops home.

Ready before the storm. Relay during it. Recover after.

~7,000 of 14,167
cell tower sites in West Bengal went dark after Cyclone Amphan, 2020

135
Philippine cities and towns lost telecoms after Typhoon Odette, 2021

VHF radio
is what Puri's officials fell back on after Cyclone Fani, 2019

Sources: The Quint, citing TAIPA (May 2020) · Philstar Interaksyon (Dec 2021) · Hindustan Times (13 May 2019)

Opportunities

How is it different from existing ideas?

- Calls and SMS need a tower. Mesh chat apps need you to know a contact. Relay needs neither.
- A structured need hops until one phone finds signal, and lands in the district control room, not a chat thread.
- Ready mode installs it before the storm, so it is already on the phone when the tower goes.

How will it solve the problem?

- Gemma turns a panicked voice note into a ~100-byte ticket offline and asks what a dispatcher would.
- Gemini agents merge duplicates, place landmarks on the map, rank needs with a reason, and draft dispatches.
- The duty officer approves; the confirmation travels back over the same hops.

USP of the proposed solution

- No signal, no literacy, no app to find: one button and a voice, in Indic and Asian languages.
- Never says “help is coming” until a human has approved it.
- Runs on phones people already own. No new hardware.

List of features offered by the solution

01
One-button voice intake
Press and say what you need. Multiple Indic and Asian languages, across dialects. Typing is the fallback.

02
Offline AI on the phone
Gemma on LiteRT-LM turns speech into a ~100-byte ticket and asks up to 3 follow-ups. Rule-based parser if the model is late.

03
Phone-to-phone mesh
Nearby Connections over BLE. Store and forward, TTL 7, ACKs. Nobody has to open an app.

04
Any phone is a gateway
One bar of signal is enough to sync tickets up and replies down over HTTPS.

05
AI triage
Embeddings merge duplicate reports; Flash + Maps turns “Pipili bus stand” into a dot; every priority has a readable reason.

06
Dispatch and sitreps
An ADK agent allocates teams within capacity and travel time; Gemini Pro writes the hourly sitrep in the district’s format.

07
Control room
Live map, priority queue, one-click human approval. The requester gets a confirmation in their language.

08
Ready and Recover modes
Before landfall: a family plan read aloud. After: a damage map by village and evidence for the state and insurers.

Process flow diagram or Use-case diagram

A structured need hops until one phone finds signal; the confirmation rides the same path home.

01
User phone
No signal. A voice note or a tap; Gemma asks 3 questions.

→

02
Relay phones
BLE mesh, TTL 7. Store and forward; nobody opens an app.

→

03
Gateway phone
One bar of signal. Uploads tickets, carries replies.

→

04
Relay cloud
Cloud Run and ADK agents. Triage, dispatch, sitrep, translate.

→

05
Control room
Live map and queue. The officer approves in one click.

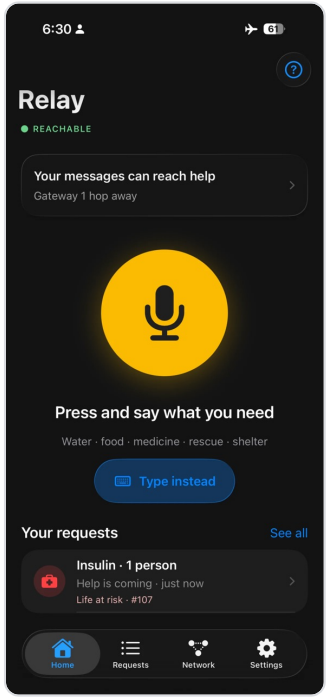
BACK THE SAME HOPS *“Boat leaving the school at 15:40. Keep her warm.”*

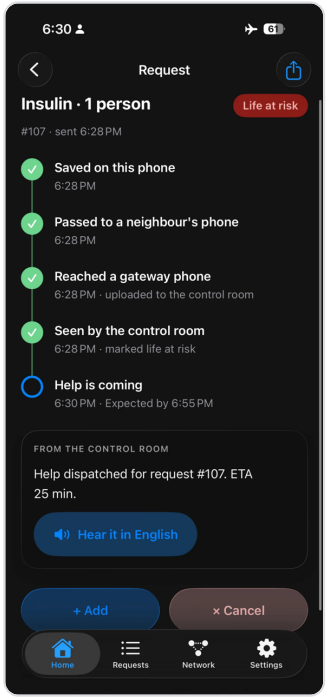
Ready before landfall: model, maps and shelters preloaded on the warning

Relay first 48 hours: ask, hop, triage, approve, confirm

Recover after the towers return: damage map and relief evidence

Wireframes/Mock diagrams of the proposed solution (optional)





Requester app

One button and a voice. No literacy, no network, no app to find.

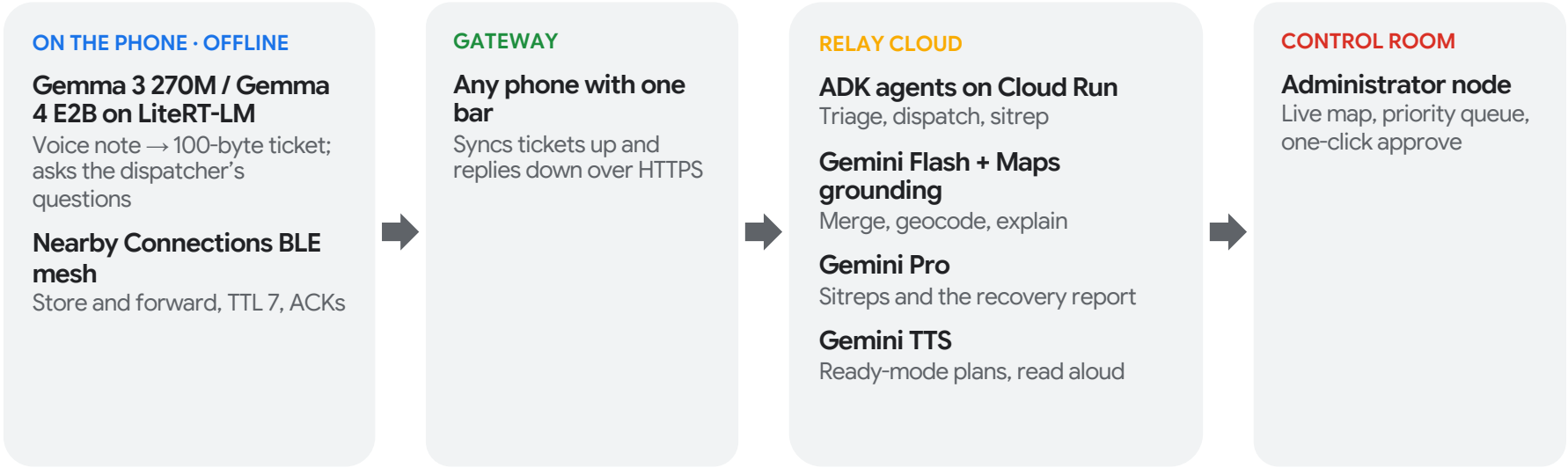
Home

Shows whether messages can reach help (“Gateway 1 hop away”), the press-and-speak button, and open requests.

Request status

Saved on this phone → passed to a neighbour → reached a gateway → seen by the control room → help is coming. Status changes only on a real ACK; the reply can be heard in the user’s language.

Architecture diagram of the proposed solution



Six Google tools, each with a job. The sixth: Gemini as the evaluation judge, checked against gold labels in English and Hindi.

Technologies to be used in the solution

ON-DEVICE

Gemma 3 270M / Gemma 4 E2B
offline intake and follow-up questions

LiteRT-LM
on-device inference runtime

Nearby Connections (BLE)
phone-to-phone mesh

Android and iOS apps
requester app and gateway

GOOGLE CLOUD

Cloud Run
serverless backend, scales with the event

Agent Development Kit (ADK)
triage, dispatch and sitrep agents

Google Maps grounding
landmarks to coordinates

GEMINI MODELS

Gemini Flash
merge, prioritise, draft, translate

Gemini Pro
sitreps and recovery reports

Gemini TTS
household plans read aloud

Gemini embeddings
duplicate-report merging

Gemini as judge
evaluates Gemma (English, Hindi)

Estimated implementation cost (optional)

< ₹1,000
cloud cost per cyclone event (est.)

< ₹2,500
per district, per season (est.)

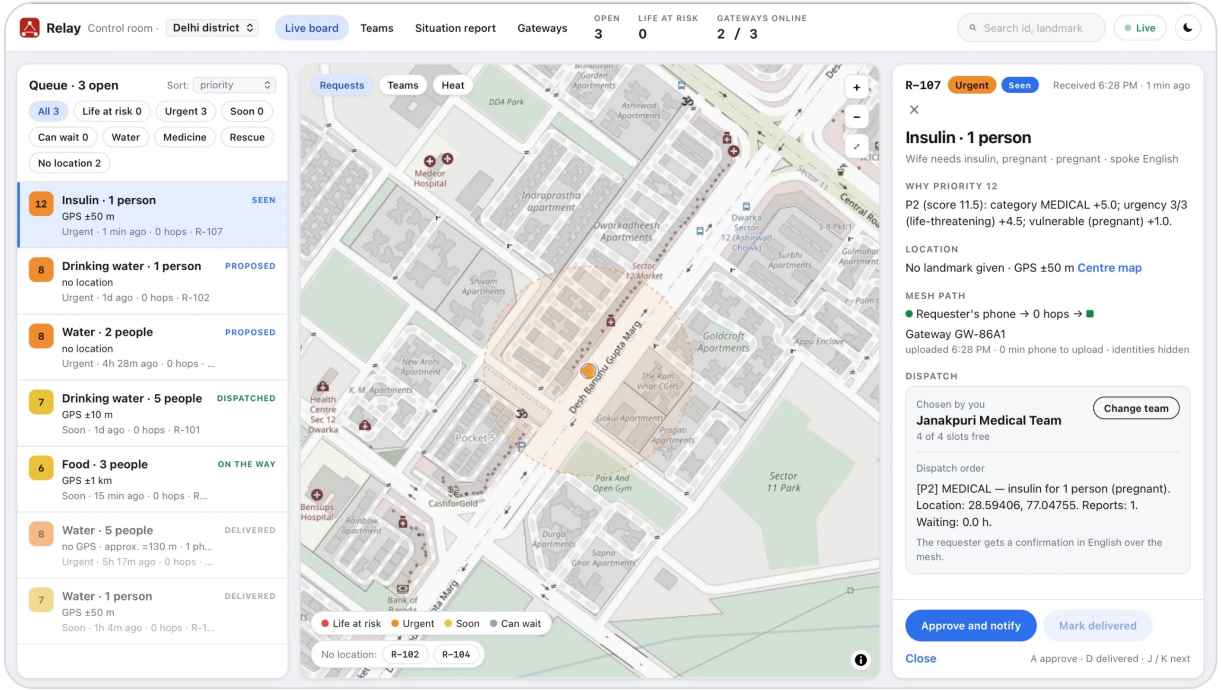
None
new hardware: runs on phones people already own

Why it stays cheap

- Intake runs on the phone: no per-request model cost in the first hours, when there is no cloud anyway.
- Tickets are ~100 bytes, so a single bar of signal carries a whole village's requests.
- Cloud Run and the agents work only when there is work; between storms the bill is close to zero.
- The mesh uses Bluetooth already in every phone: no towers, radios or satellite kit to buy.

Estimate, not a quote: ~1,000 tickets over 48 h, hourly sitreps, Rs 85/USD, Gemini list prices (assumed). Cloud Run, Firestore and Pub/Sub stay inside free tiers.

Snapshots of the prototype



Control room

Live at relay.harshvs.com

Queue

Sorted by priority, with filters for life at risk, urgent, water, medicine and rescue.

Map

Requests, teams and a heat view.

Detail

Why this priority, location accuracy, the mesh path and a drafted dispatch. Approve and notify in one click.

Prototype Performance report/Benchmarking

≈100 B
per ticket on the wire

1–3
hops, phone to gateway

≈ 2.5 s
typed send to control room

≈ 5,700 ms
Gemma intake turn (p50)

85%
category accuracy, English + Hindi

40 → 35
duplicate reports merged

Demo scenario: a synthetic, labelled forty-ticket cyclone night. 40 tickets in, 6 dispatches out (one per team) plus an hourly sitrep, with 1 human click to approve each.

Measured: tap-to-ticket median of 4 runs on a Galaxy A52 to relay.harshvs.com (typed, 0 hops, Wi-Fi). Gemma 4 E2B on-device, p50 over 34 clips; category accuracy 29/34 on audio. Duplicates: 40-ticket seed, 8 reports in 3 clusters.

RUNNING TODAY

- BLE mesh, store-and-forward with ACKs
- Gateway sync and localised confirmations
- Control room: map, queue, reasons, approve

ADDED FOR THE CUP

- Gemma intake conversation, offline
- Control-room agents: merge, dispatch, sitrep

ON RECORDED DATA, LABELLED

- Photo tickets that travel as tags
- Ready mode household plans

Additional Details/Future Development (if any)

NOW · PILOT

One Odisha district

- District control room on Relay
- Ready mode before the warning
- Baseline vs Relay: time to help

NEXT · STATE

Every coastal district

- SMS and IVR for feature phones
- Per-device signing
- Multi-district control rooms

THEN · JAPAC

The cyclone belt

- Bangladesh, Philippines, Vietnam
- Data-mule mode between islands
- Rumour checks before broadcast

What a district pilot measures

- Minutes from ask to dispatch
- Share of tickets with a usable location
- Duplicate reports merged before dispatch
- Needs confirmed closed, by village
- Households reached, in the recovery report

Why it matters

119M people are exposed to tropical cyclones each year; 85%+ of expected annual cyclone deaths fall in Bangladesh and India.

Source: PreventionWeb (UNDRR)

Provide links to your:

- **GitHub Public Repository:** https://github.com/harshvs99/ai-builder-cup
- **Demo Video Link (3 Minutes)** drive.google.com/file/d/15octzsyQ2cygVJEjCYpAf6gC18axCi2F/view
- **Final Product Link:** relay.harshvs.com (plus the accompanying Android and iOS apps)

Google Cloud H2S

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Build in JAPAC.
Build for the world.

Thank you

